MODULE 6: CONTEMPORARY TOPICS IN COMPUTATIONAL CHEMISTRY & MATERIALS (CO: 2)

Advanced Contemporary Textbook: Machine Learning in Material Science, Graph Neural Networks, Solid-State Batteries & Solved VIT FAT Question Bank

Aligned with Official VIT BACHY105 Applied Chemistry Course Syllabus & VIT FAT Contemporary Topics

Syllabus Scope & Competency Checklist (Module 6):

- **Contemporary Topics in Computational Chemistry:** Evolution of quantum simulations; Crystal Graph Convolutional Neural Networks (CGCNN); Machine Learning force fields (DeepMD, M3GNet) replacing classical empirical potentials; generative models (Diffusion, VAEs) for de novo crystalline material design.
- **Contemporary Topics in Energy Devices & Functional Materials:** Beyond-lithium technologies (All-Solid-State Lithium Batteries ASSBs, Sodium-Ion Batteries); solid electrolytes ($Li_7La_3Zr_2O_{12}$ LLZO garnet, $Li_{10}GeP_2S_{12}$ LGPS thiophosphate); high-entropy alloys (HEAs); lead-free double perovskite photovoltaics ($Cs_2AgBiBr_6$).

1. Machine Learning & Graph Neural Networks in Materials Discovery

1.1 The Materials Genome & Crystal Graph Representations

Traditional materials discovery relied on trial-and-error synthesis taking $15-20$ years from laboratory discovery to commercial deployment. By encoding crystal structures into mathematical graphs $G=(V,E)$:

- **Nodes (V):** Atoms encoded by feature vectors (electronegativity, ionization energy, covalent radius, oxidation state).
- **Edges (E):** Interatomic bonds defined by Voronoi tessellation and Euclidean distances.

Crystal Graph Convolutional Neural Networks (CGCNN) perform message passing across atomic neighbors:

$$\mathbf{v}_i^{(t+1)} = \mathbf{v}_i^{(t)} + \sum_{j \in N(\mathrm{i})} g\left(\mathbf{v}_i^{(t)}, \mathbf{v}_j^{(t)}, \boldsymbol{e}_{ij}\right)$$

enabling direct, translation- and rotation-invariant predictions of thermodynamic stability (energy above convex hull E_{hull}) and electronic properties in milliseconds with DFT-level precision.

1.2 All-Solid-State Batteries (ASSBs) & Superionic Conductors

Solid-state batteries replace flammable organic liquid electrolytes with non-flammable solid ceramic or polymer electrolytes:

- **Superionic Solid Electrolytes:**
  1. **Oxide Garnet ($\mathbf{Li_7La_3Zr_2O_{12}}$ LLZO):** Excellent chemical stability against metallic lithium anodes, ionic conductivity $\approx 10^{-3}$ S/cm at room temperature.
  2. **Thiophosphates ($\mathbf{Li_{10}GeP_2S_{12}}$ LGPS):** Possesses interconnected 1D diffusion pathways providing extraordinary lithium ionic conductivity (1.2×10^{-2} S/cm, rivaling liquid electrolytes).

High-throughput ML screening models screen millions of chemical stoichiometries to identify candidates with wide electrochemical stability windows (> 4.5 V) and high shear moduli to mechanically suppress lithium dendrite growth.

VIT FAT Solved Question Bank: Contemporary Exam Problem

FAT Problem 1: Solid-State Battery Electrolyte Ionic Conductivity & Arrhenius Transport
Question (10 Marks): An advanced solid electrolyte material ($Li_7La_3Zr_2O_{12}$ LLZO) has an ionic conductivity of $\sigma_1 = 1.05 \times 10^{-3}$ S/cm at $T_1 = 298$ K and $\sigma_2 = 4.80 \times 10^{-3}$ S/cm at $T_2 = 348$ K. ("i") Assuming Arrhenius ion-hopping kinetics ($\sigma T = Ae^{-\frac{E_a}{k_B T}}$), calculate the activation energy E_a for lithium ion transport in electron-volts (eV). ("ii") Compute the pre-exponential factor A. ("iii") Determine the predicted ionic conductivity at an arctic operating temperature of $-20°C$ (253 K).

Complete Step-by-Step Solution:
Step 1: Calculate Activation Energy E_a:

$$\ln\left(\frac{\sigma_2 T_2}{\sigma_1 T_1}\right) = -\frac{E_a}{k_B}\left[\frac{1}{T_2} - \frac{1}{T_1}\right] = \frac{E_a}{k_B}\left[\frac{T_2 - T_1}{T_1 T_2}\right]$$

$$\sigma_1 T_1 = (1.05 \times 10^{-3}) \times 298 = 0.3129 \text{ S K/cm}$$

$$\sigma_2 T_2 = (4.80 \times 10^{-3}) \times 348 = 1.6704 \text{ S K/cm}$$

$$\frac{\sigma_2 T_2}{\sigma_1 T_1} = \frac{1.6704}{0.3129} = 5.3384$$

$$\ln(5.3384) = 1.6749$$

$$\frac{T_2 - T_1}{T_1 T_2} = \frac{348 - 298}{298 \times 348} = \frac{50}{103704} = 4.8214 \times 10^{-4} \text{ K}^{-1}$$

With $k_B = 8.6173 \times 10^{-5}$ eV/K:

$$1.6749 = \frac{E_a \times 4.8214 \times 10^{-4}}{8.6173 \times 10^{-5}} = E_a \times 5.5950$$

$$E_a = \frac{1.6749}{5.5950} = 0.2994 \text{ eV} \approx 0.30 \text{ eV}$$

Step 2: Pre-Exponential Factor A:

$$\ln A = \ln(\sigma_1 T_1) + \frac{E_a}{k_B T_1} = \ln(0.3129) + \frac{0.2994}{(8.6173 \times 10^{-5}) \times 298} = -1.1619 + \frac{0.2994}{0.02568} = -1.1619 + 11.659 = 10.497$$

$$A = e^{10.497} = 36207 \text{ S K/cm}$$

Step 3: Conductivity at $-20°C$ ($T_3 = 253$ K):

$$\sigma_3 T_3 = Ae^{-\frac{E_a}{k_B T_3}}$$

$$\frac{E_a}{k_B T_3} = \frac{0.2994}{(8.6173 \times 10^{-5}) \times 253} = \frac{0.2994}{0.02180} = 13.733$$

$$\sigma_3 \times 253 = 36207 \times e^{-13.733} = 36207 \times (1.086 \times 10^{-6}) = 0.03932$$

$$\sigma_3 = \frac{0.03932}{253} = 1.554 \times 10^{-4} \text{ S/cm}$$

Boxed Final Answers:

$$E_a = 0.30 \text{ eV}, \quad A = 3.62 \times 10^4 \text{ S K/cm}, \quad \sigma(-20°C) = 1.55 \times 10^{-4} \text{ S/cm}.$$